Hybrid Classical-Quantum Autoencoder for Image Denoising

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1 Introduction

Image denoising aims to recover a clean signal from a noisy observation [3], as noise arising from sensor or acquisition imperfections degrades the performance of downstream vision systems [8, 38]. Deep learning approaches, such as convolutional denoisers and autoencoders, now dominate this task, learning robust latent representations that separate signal from noise [38, 31, 32].

In parallel, quantum information processing [22, 36] has motivated the development of Quantum Machine Learning (QML) models that exploit superposition and entanglement for data processing [10, 2, 9, 6, 17]. Many near-term implementations rely on parametrized quantum circuits (PQCs) trained within hybrid classical–quantum optimisation loops, commonly known as Variational Quantum Algorithms (VQAs) [24, 4, 25, 23]. Such hybrid models are attractive in the NISQ era due to their hardware compatibility and potential representational benefits.

In this work, we study a hybrid denoising autoencoder in which a classical encoder maps images to a compact latent space, which is then processed by a variational quantum circuit (VQC) before being decoded back to the image domain. This architecture leverages the natural bottleneck structure of autoencoders to restrict the quantum component to a low-dimensional representation, making the approach compatible with current devices.

2 Related Work

Image denoising has long been studied in classical image processing and computer vision. Early approaches relied on handcrafted priors over image statistics, such as Non-Local Means [3] and BM3D [8]. With deep learning, data-driven pipelines have become dominant: convolutional networks like DnCNN learn residual mappings from noisy to clean images across noise levels [38], while denoising autoencoders learn compact latent representations that suppress noise while preserving structure [31, 32]. This latent-space view is especially appealing for hybrid classical-quantum models, where quantum resources are limited.

Quantum Machine Learning (QML) integrates quantum information processing with learning-based models to explore new representational paradigms [2]. In the NISQ era, most practical methods use hybrid quantum–classical optimisation with PQCs, known as Variational Quantum Circuits (VQCs) [24, 4]. These models embed classical data into quantum states and optimise circuit parameters in classical loops [25, 23], making them well-suited to near-term hardware.

Hybrid classical–quantum models have been explored in several computer-vision tasks. In image denoising, hybrid autoencoder architectures replacing the latent space with a variational quantum

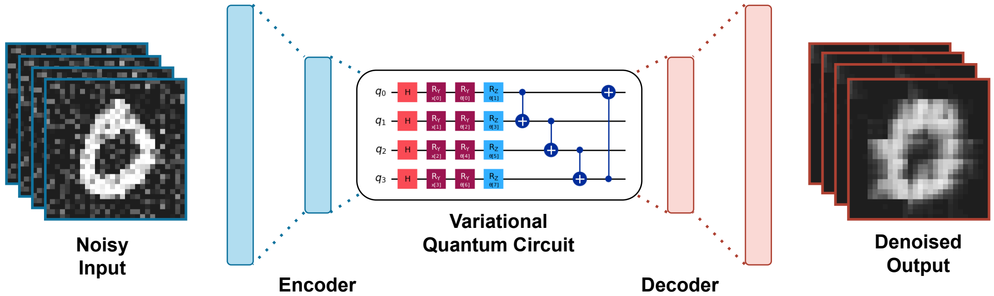


Figure 1: Architecture of our hybrid classical–quantum variational autoencoder, where the classical encoder and decoder are connected through a quantum variational circuit acting as the latent space.

circuit have been proposed and evaluated on standard benchmarks [16]. Related approaches include transfer-learning pipelines combining classical feature extractors with variational quantum classifiers [20], quantum feature-generation layers [13], and variational quantum formulations for image segmentation and generative modeling [29, 28]. More recent works investigate quantum-enhanced continuous visual representations and hybrid architectures for image synthesis and representation [39, 33, 7].

Despite promising experimental demonstrations [1, 11], VQAs face scalability challenges such as barren plateaus, which lead to vanishing gradients as system size increases [21, 18]. Nevertheless, ongoing research on structured circuit designs and improved training strategies continues to support the potential of hybrid quantum–classical models for practical image processing tasks [30, 23, 14].

3 Methodology

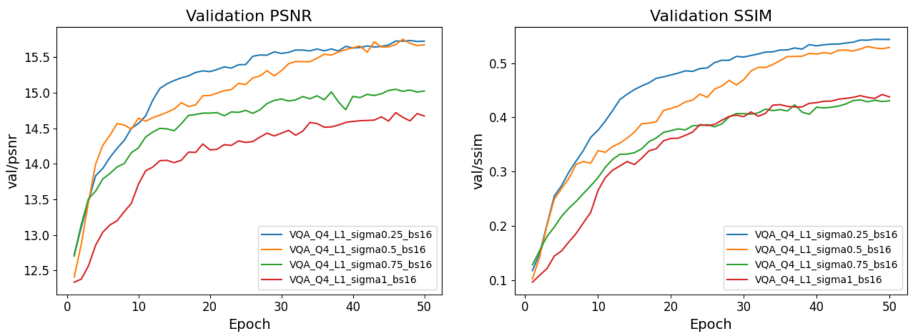
Our model implements a hybrid classical-quantum autoencoder, where the latent representation is processed by a variational quantum circuit. An overview of the architecture is presented in Fig. 1. This structure is analogous to prior hybrid autoencoder approaches, including the one proposed in reference [16].

Given an input image $x \in \mathbb{R}^d$, a classical convolutional encoder E maps x to a latent representation $\mathbf{z} = E(x)$. The latent vector $\mathbf{z}$ is normalized and mapped to n qubits using angle encoding. For each component z_i, a single-qubit rotation gate encodes the value so that the encoded n-qubit state factorizes as

$$|\psi_{\mathbf{z}}\rangle = \bigotimes_{i=1}^{n} R_Y(z_i)\,|0\rangle. \tag{1}$$

The choice of angle encoding provides a simple and hardware-efficient mapping of classical latent variables into quantum states.

On top of this encoding we apply a trainable ansatz consisting of repeated layers of single-qubit rotations followed by linear entanglement with an optional ring closure. For L layers, the variational



(a) (b)

Figure 2: Evolution of the (a) PSNR and (b) SSIM for the quantum model with different levels of noise.

unitary is

$$U(\boldsymbol{\phi}) = \prod_{\ell=1}^{L} \left[\left(\prod_{j=1}^{n} R_Y(\phi_{j,1}^{(\ell)}) R_Z(\phi_{j,2}^{(\ell)}) \right) \prod_{j=1}^{n-1} \mathrm{CNOT}(j \to j+1) \right], \tag{2}$$

with an additional $\mathrm{CNOT}(n \to 1)$ when ring coupling is used. The resulting quantum state is

$$|\psi(\mathbf{z}, \boldsymbol{\phi})\rangle = U(\boldsymbol{\phi}) \, |\psi_{\mathbf{z}}\rangle. \tag{3}$$

We employ Pauli-Z measurements on each qubit, and the corresponding expectation values are returned to the classical pipeline as the quantum-processed latent representation $\tilde{\mathbf{z}}$.

The output of the quantum model, denoted $\tilde{\mathbf{z}} \in \mathbb{R}^m$, is passed to the classical decoder D, yielding the reconstruction $\hat{x} = D(\tilde{\mathbf{z}})$. The network is trained end-to-end by minimizing the mean–squared error (MSE) reconstruction loss

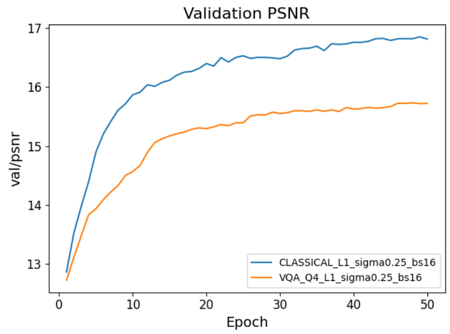
$$\mathcal{L}_{\mathrm{MSE}}(x, \hat{x}) = \frac{1}{d} \, \|x - \hat{x}\|_2^2, \tag{4}$$

where d is the dimensionality of the input.

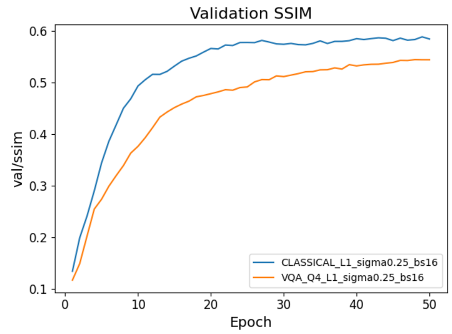
This design balances expressivity and trainability: single-qubit rotations cover SU(2) locally, while the linear entanglement pattern introduces multi-qubit correlations without creating highly entangled, hard-to-optimize states. The parameter count grows linearly with qubits and layers, making it suitable for NISQ hardware. Stacking layers yields a tunable hierarchy of nonlinear feature maps in the latent space, and Pauli-Z readout fits naturally with the gate set, providing structured measurement statistics.

4 Results

To evaluate the suitability of variational quantum circuits for image denoising, a set of experiments was designed to analyse both training behaviour and denoising performance. The optimisation process exhibits stable convergence across different batch sizes and parameter initialisations, indicating reliable training dynamics. All experiments were conducted on the MNIST dataset [19], composed



(a)



(b)

Figure 3: Evolution of the (a) PSNR and (b) SSIM for the classical model with different levels of noise.

of 70,000 28×28 grayscale images of handwritten digits, split into 60,000 training and 10,000 test samples. For our experiments, we are constraining the dataset to 2,000 samples of numbers 0 and 1 for training, and 200 samples of these numbers for validation due to computational limitations.

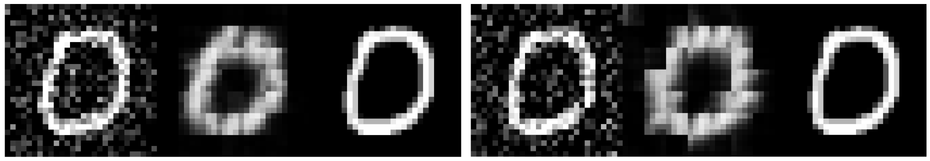
All experiments were conducted with the same hardware configuration on the PIC computing cluster [1], using a single NVIDIA RTX 4000 Ada Generation GPU. The node provided 32 CPU cores (64 logical threads) and Python 3.12 was used throughout the experiments.

A detailed analysis of training convergence and sensitivity is provided in Appendix B.An ablation study on architectural hyperparameters shows that, within the explored range, variations in the number of qubits and circuit layers have a limited impact on performance. In particular, shallow circuits with few qubits already achieve competitive results, suggesting that the classical encoder captures most of the relevant information. A more detailed analysis is provided in Appendix C. Training dynamics were monitored through the evolution of the loss function, and performance was assessed using standard image reconstruction metrics (PSNR [15, 34], SSIM [35], MSE [34], MAE [37]). Complete quantitative results are provided in a summary table available online[2]. In the following, we focus on the denoising performance results.

Noise level. The robustness of the VQA-based model was evaluated under different noise levels applied to the input images. As shown in Figure 2, denoising performance degrades as noise increases. While the model effectively reduces moderate noise levels, its performance decreases in more challenging scenarios, becoming harder to correct with a fixed circuit structure. Nevertheless, reasonable reconstruction quality is maintained for intermediate noise levels, indicating a degree of robustness of the proposed approach.

The best VQA-based configuration was compared with a purely classical baseline trained under identical conditions. As shown in Figure 3, the classical model achieves consistently strong performance, reflecting the maturity of deep learning-based denoising techniques. Nevertheless, the VQA-based model attains comparable performance in low to intermediate noise regimes while using fewer than 3,000 trainable parameters, compared to 3,709 parameters in the classical baseline. This

[1] Cluster located at the campus of the Universitat Autònoma de Barcelona, available online here.
[2] All numerical results are available at Google Sheets.



(a) (b)

Figure 4: Examples of denoising obtained with (a) the classical model and (b) the hybrid classical-quantum model.

suggests that the hybrid classical–quantum architecture is able to capture relevant image features with a more compact parameterisation. At higher noise levels, the performance gap increases due to the limited quantum resources, circuit depth constraints, and optimisation noise.

Additionally, a qualitative comparison is shown in Figure 4. While both models suppress a significant amount of noise, the classical approach produces cleaner reconstructions with sharper contours and fewer artefacts. The VQA-based model nevertheless preserves the global structure of the images, demonstrating its ability to capture relevant features despite limited quantum resources.

5 Conclusions & Discussion

Classical deep learning models benefit from decades of algorithmic development and are typically optimised for image processing tasks in high-dimensional spaces, while the VQA-based model operates under constraints imposed by the limited number of qubits, circuit depth, and the presence of noise in the variational optimisation process.

Nevertheless, the fact that the quantum model achieves competitive performance in certain regimes is a promising result, suggesting that variational quantum circuits can capture relevant structural information for denoising tasks, even with relatively simple circuit architectures and limited resources. These results highlight the potential of hybrid quantum–classical approaches, particularly as quantum hardware and training techniques continue to improve.

A natural direction for future work is to extend this proof-of-concept to larger-scale settings and more challenging noise models. This includes exploring deeper and more expressive variational ansätze, improved data-encoding strategies, adaptive or layer-wise training protocols, and error-mitigation techniques to stabilise optimisation under realistic noise. Evaluating performance on higher-resolution datasets, real-world degradation processes, and alternative hybrid configurations (e.g., quantum feature extractors or decoder-level quantum modules) would further clarify where quantum components genuinely add value. Ultimately, this work should be viewed as an early step toward understanding the role of quantum latent representations in vision pipelines, with the expectation that both theory and hardware advances will be required to fully realise their potential.

Code Availability

The coding and data material of this project is available in our public GitHub repository [3]. The MNIST Dataset used in our experiments is also available online.

[3] You can access the repository in the following link: https://github.com/martabenavente/VQE-Image-denoising

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A Introduction to Quantum Machine Learning

Quantum Machine Learning (QML) represents an emerging interdisciplinary field that integrates principles of quantum computing with machine learning methodologies to leverage quantum phenomena such as superposition and entanglement for solving complex computational problems [2, 27]. This appendix introduces fundamental quantum computing concepts integral to QML, specifically targeted at the computer science audience, with mathematical formulations and common applications to bridge understanding between classical and quantum paradigms.

Dirac notation, employing the symbol $|\psi\rangle$ known as a "ket", provides a convenient way to denote quantum states without referring to the particular function used to represent them [22]. This mathematical formalism, developed by Paul Dirac, forms the foundation for expressing quantum mechanical operations and is essential for understanding quantum machine learning algorithms. The notation consists of three fundamental components. Kets represent quantum states $|\psi\rangle$ which correspond to vectors in a complex Hilbert space $\mathcal{H}$. Mathematically, a ket can be expressed as

$$|\psi\rangle = \sum_i c_i |i\rangle \,,$$

where $\{c_i\}$ are complex coefficients and $\{|i\rangle\}$ form an orthonormal basis of the Hilbert space. Bras are defined as the complex conjugate transpose of kets. For a ket $|\psi\rangle = \sum_i c_i |i\rangle$, the corresponding bra is

$$\langle\psi| = \sum_i c_i^* \langle i| \,,$$

where c_i^* denotes the complex conjugate of c_i. The bra-ket notation also defines scalar products between quantum states. The inner product between two states $|\phi\rangle = \sum_i d_i |i\rangle$ and $|\psi\rangle = \sum_j c_j |j\rangle$ is given by

$$\langle\phi|\psi\rangle = \sum_{i,j} d_i^* c_j \langle i|j\rangle = \sum_i d_i^* c_i,$$

where the orthonormality condition $\langle i|j\rangle = \delta_{ij}$ has been used. Finally, the notation includes outer products, which represent operators. The outer product of two states $|\psi\rangle$ and $\langle\phi|$ is written as

$$|\psi\rangle\langle\phi| = \sum_{i,j} c_i d_j^* |i\rangle\langle j| \,.$$

These operators are widely used in quantum mechanics and quantum computing for constructing density matrices and projection operators.

A qubit, or quantum bit, is the fundamental unit of quantum information. Unlike a classical bit, which is represented as either 0 or 1, a qubit exists in a superposition of the basis states $|0\rangle$ and $|1\rangle$: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle\,, \quad \text{where} \quad |\alpha|^2 + |\beta|^2 = 1$. Here, $\alpha, \beta \in \mathbb{C}$ are complex amplitudes that determine the probabilities of measuring $|0\rangle$ and $|1\rangle$, respectively. This probabilistic nature lies at the heart of quantum computation [22].

Quantum measurement collapses a quantum state $|\psi\rangle$ into one of its basis states. The probability of measuring a state $|i\rangle$ is given by

$$P(|i\rangle) = |\langle i|\psi\rangle|^2. \tag{5}$$

Measurement outcomes are fundamental to extracting classical information from a quantum system and are central to the functioning of hybrid quantum-classical models.

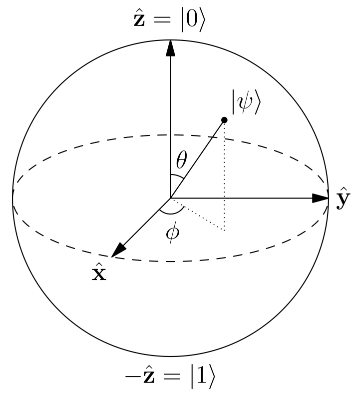


Figure 5: The Bloch sphere representation of a qubit state. The angles θ and ϕ define the position of the state vector on the sphere, encoding the superposition and relative phase of the basis states $|0\rangle$ and $|1\rangle$.

Superposition allows a single qubit to represent multiple states simultaneously. For example, a superposition state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ represents an equal probability of measuring either $|0\rangle$ or $|1\rangle$, where each outcome has probability $|\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$. In systems with multiple qubits, entanglement emerges as a key quantum phenomenon, where the quantum state of one qubit is dependent on another, even if spatially separated

$$|\psi_{\text{entangled}}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle). \tag{6}$$

Such correlations cannot be explained by classical physics and are pivotal in QML algorithms and methods.

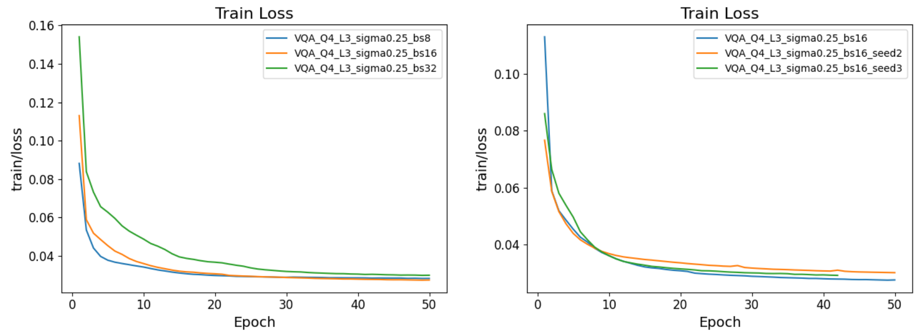
The Bloch sphere provides a geometric representation of a qubit as a point on a unit sphere in 3D space. Any qubit state $|\psi\rangle = \cos(\theta/2)\,|0\rangle + e^{i\phi}\sin(\theta/2)\,|1\rangle$ can be represented on the sphere through angles θ (polar) and ϕ (azimuthal). This visualization succinctly captures superposition and phase relationships and is widely used to illustrate quantum state evolution under quantum gates (see Figure 5).

Quantum gates are unitary operations that manipulate qubits, analogous to logical gates in classical computation. Common single-qubit gates include

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

The X-gate is a quantum analog of the NOT gate, while the Hadamard (H) gate places a qubit in superposition. Multi-qubit gates like the controlled-NOT (CNOT) enable entanglement by conditioning one qubit's state on another

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$



(a) (b)

Figure 6: Evolution of the loss function during training of the quantum model modifying (a) the size of batches and (b) the initialisation seed.

Feature maps encode classical data $\mathbf{x} \in \mathbb{R}^n$ into quantum states. The encoding is achieved through unitary transformations $U(\mathbf{x})$ that map data into a high-dimensional quantum Hilbert space. This process is formally represented as

$$|\phi(\mathbf{x})\rangle = U(\mathbf{x}) |0\rangle . \tag{7}$$

For example, *amplitude encoding* embeds $\mathbf{x}$ into the amplitudes of a quantum state, ensuring that $\|\mathbf{x}\|^2 = 1$. Other commonly used techniques in quantum machine learning include *basis encoding*, where each component of $\mathbf{x}$ is mapped to computational basis states, and *angle embedding*, in which the components of $\mathbf{x}$ are used as rotation angles in parameterized quantum gates (e.g., $R_y(x_i)$ or $R_z(x_i)$). Angle embedding is particularly hardware-friendly and suitable for near-term quantum devices [26, 12].

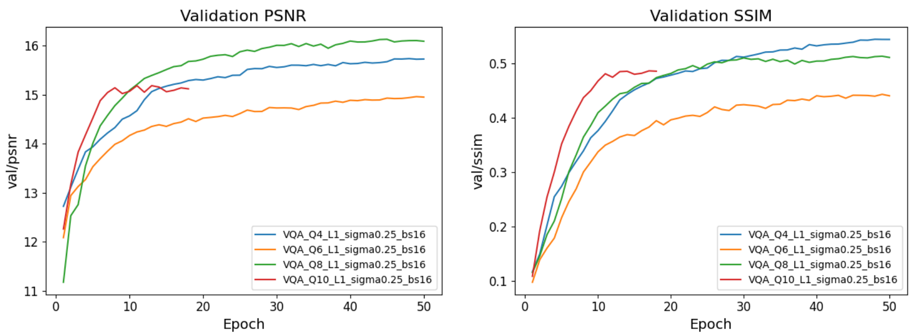
An ansatz in quantum computing refers to a parameterized quantum circuit used to approximate solutions to optimisation problems in QML. These circuits are constructed as layers of gates, with each gate having trainable parameters. The full circuit can be expressed as

$$U(\boldsymbol{\theta}) = \prod_{l=1}^{L} U_l(\boldsymbol{\theta}_l), \tag{8}$$

where the parameters $\boldsymbol{\theta}$ are optimised using classical algorithms to minimize a defined cost function. Common ansatz designs include hardware-efficient and problem-inspired ansätze [6].

B Additional Training Analysis

To complement the main results, this appendix provides a detailed analysis of the training behaviour of the proposed variational quantum model. The experiments were designed to evaluate both convergence properties and robustness of the optimisation process, while keeping the remaining hyperparameters fixed for fair comparison across configurations.



(a) (b)

Figure 7: Evolution of the (a) PSNR and (b) SSIM for the quantum model with different number of qubits.

B.1 Training & Convergence

The convergence behaviour of the optimisation process was analysed through the evolution of the loss function and the variability between independent runs.

Figure 6 shows a progressive decrease of the loss during training, indicating that the model successfully adapts its parameters to improve image reconstruction. In most configurations, the loss stabilises after a moderate number of epochs, suggesting convergence of the optimisation process. Variability between runs is more pronounced during the early training stages, as expected in variational quantum models due to parameter initialisation and sampling noise [5].

Despite this initial variability, all runs converge towards similar loss values, and no systematic divergences or unstable behaviours are observed.

B.2 Sensitivity & Stability

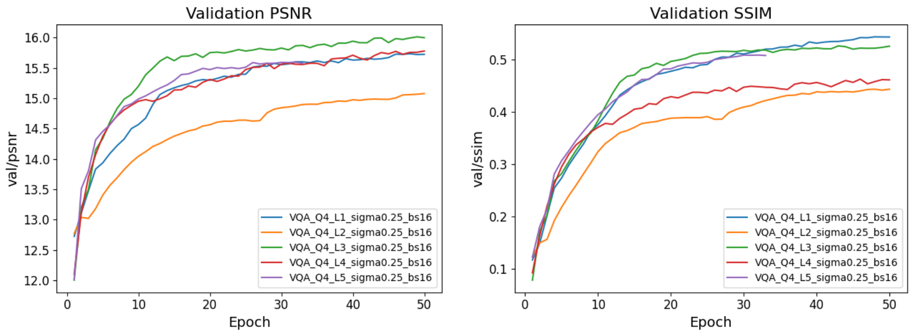
The robustness and reproducibility of the training process were evaluated by analysing the impact of batch size and parameter initialisation, while keeping the remaining experimental settings fixed.

Different batch sizes were tested, as shown in Figure 6 (a). Larger batch sizes result in slower convergence and slightly higher final loss values, whereas smaller and intermediate batch sizes converge faster with comparable performance. No significant instabilities were observed across the tested configurations.

The sensitivity to initial conditions was assessed by varying the random seed. As illustrated in Figure 6 (b), although early-stage variability is present, all runs converge to similar loss values, confirming the robustness of the optimisation process with respect to parameter initialisation.

C Additional Ablation Study

Number of qubits. They determine the dimensionality of the quantum state used to encode the latent representation of the image (as explained in Sec. 3), and is therefore directly related to the representational capacity of the quantum model. As shown in Figure 7, while an increase from 4 to 8 qubits leads to a moderate improvement in PSNR, the trend is not consistent across



(a) (b)

Figure 8: Evolution of the (a) PSNR and (b) SSIM for the quantum model with different number of layers.

all configurations, as the 6-qubit model performs noticeably worse. Moreover, in terms of SSIM, the 4-qubit model achieves the best overall performance, suggesting that increasing the number of qubits does not necessarily translate into better perceptual quality. Overall, these results indicate that, for the considered architecture and latent dimensionality, increasing the number of qubits has a limited practical impact on denoising performance.

Number of layers. Determinates the depth of the variational circuit, which plays a crucial role in determining the expressivity of the model. We can appreciate in Figure 8 that an increase in the number of layers does not lead to a consistent improvement in denoising performance. In terms of SSIM, the single-layer circuit achieves the best overall performance, closely followed by the 3-layer configuration, while deeper circuits do not provide additional perceptual benefits. For PSNR, the 3-layer model attains slightly higher values, which is expected given the use of a pixel-wise reconstruction loss closely related to PSNR. Overall, these results suggest that shallow circuits are sufficient to process the latent representations produced by the classical encoder.